Zhuochun Li

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Education

University of Pittsburgh Pittsburgh, USA
Ph.D. in Information Science Sep 2023 – Present
Master of Science in Information Science, GPA: 3.9/4.0 Jan 2021 – Dec 2022
Xi'an University of Technology Sep 2016 – Jul 2020
Bachelor of Engineering in Computer Science Xi'an, China

Profile

Highly motivated Ph.D. student in Information Science with research interests in **NLP** and **Machine Learning**. Proficient in Python, Machine Learning, and advanced NLP methods. Research focuses on **LLM reasoning**, **post-training**, and **knowledge distillation**. Completed **Applied Scientist Intern** at **Amazon**; incoming **Applied Scientist Intern** at **Microsoft** (Summer 2026). Published research in top venues such as **SIGIR, EMNLP, ACL,** and **ICLR**.

Selected Publications [Google Scholar]

Li, Zhuochun, Zhang Yong, Ming Li. Rethinking LLM-as-a-Judge: Representation-as-a-Judge with Small Language Models via Semantic Capacity Asymmetry. ***ICLR 2026***

César Guerra-Solano, **Zhuochun Li,** Xiang Lorraine Li. Think Globally, Group Locally: Evaluating LLMs Using Multi-Lingual Word Grouping Games. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing **(EMNLP)**,* pp. 20155-20176. 2025.

Li, Zhuochun, Yuelyu Ji, Rui Meng, Daqing He. Learning from Committee: Reasoning Distillation from a Mixture of Teachers with Peer-Review. In *Findings of the Association for Computational Linguistics: **ACL 2025**,* pages 4190–4205, Vienna, Austria. Association for Computational Linguistics.

Ji, Yuelyu, **Zhuochun Li**, Rui Meng, Daqing He. Reason-to-Rank: Distilling Direct and Comparative Reasoning from Large Language Models for Document Reranking. In *Proceedings of the 48th International ACM **SIGIR** Conference on Research and Development in Information Retrieval,* pp. 2320-2329. 2025.

Ji, Yuelyu, **Zhuochun Li**, Rui Meng, Sonish Sivarajkumar, Yanshan Wang, Zeshui Yu, Hui Ji, Yushui Han, Hanyu Zeng, and Daqing He. "RAG-RLRC-LaySum at BioLaySumm: Integrating Retrieval-Augmented Generation and Readability Control for Layman Summarization of Biomedical Texts." In *Proceedings of the 23rd Workshop on Biomedical Natural Language Processing: **ACL 2024***, pp. 810–817, Bangkok, Thailand. Association for Computational Linguistics.

Li, Zhuochun, Bo Xie, Robin Hilsabeck, Alyssa Aguirre, Ning Zou, Zhimeng Luo, and Daqing He. "Effects of Different Prompts on the Quality of GPT-4 Responses to Dementia Care Questions." In *2024 12th IEEE International Conference on Health Informatics (ICHI).* IEEE, 2024.

Li, Zhuochun, Khushboo Thaker, and Daqing He. "SiaKey: A Method for Improving Few-shot Learning with Clinical Domain Information." In *2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 1-4. IEEE, 2023.

Li, Zhuochun, Wang, Zhixiao, Wenyao Yan, Min Huang, Qinyuan Fan, and Xin Wang. "Domestic Violence Crisis Recognition Method based on Bi-LSTM+ Attention." In *2022 8th Annual International Conference on Network and Information Systems for Computers (ICNISC)*, pp. 569-575. IEEE, 2022.

Academic Experience

PhD Preliminary Exam Project, University of Pittsburgh Mar 2024 – Mar 2025

Enhance the reasoning ability of student LM via knowledge distillation from multiple LLMs. *Pittsburgh, U.S.*

- Introduced a novel Fault-Aware Distillation via Peer-Review (FAIR) approach that enables student LLMs to better acquire reasoning skills from multiple teacher LLMs: ChatGPT, Gemini, and Mistral.
- Helped student LM learn not only from the gold-standard rationale but also from feedback on their own mistakes via instruction tuning through a simulated peer-review process between teacher LLMs.

- Demonstrated the effectiveness of our method across comprehensive experiments and analysis on mathematical(GSM8K, SVAMP), commonsense(StrategyQA), and logical(LogiQA) reasoning tasks.

Research Assistant, iRiS Lab at University of Pittsburgh May 2022 – May 2023

Improve Few-shot Learning with Clinical Domain Information. *Pittsburgh, U.S.*

- Studied few-shot learning methods in the application of text classification and recommendation systems.
- Achieved average accuracy of over 60% by 10-shot training on Siamese Networks via triple loss function, which is comparable to the BERT performance using the whole dataset on this task.

Computer Vision - 16-720A, Carnegie Mellon University Jan 2022 - Apr 2022

Complete the Computer Vision course with grade A taught by Professor David Held of CMU *Pittsburgh, U.S.*

- Learned technologies about Spatial Pyramid Matching, Planar Homographies and Lucas-Kanade Tracking.
- Contributed to tasks about 3D Reconstruction, Neural Networks for Recognition and Photometric Stereo.
- Comprehended the cutting-edge deep learning models such as GAN, VAE and Transformers.

Undergraduate Thesis Jan 2020 - Jul 2020

Implementation of Domestic Violence Crisis Recognition Method Based on Deep Learning *Xi'an, China*

- Collected 1654 posts related to Domestic violence and built 50-dimensional word vectors via Word2Vec.
- Constructed CNN, RNN, LSTM, Bi-LSTM+self-Attention neural network models to accomplish text categorization task, Bi-LSTM+self-Attention model had the best performance, with an accuracy rate of 90.22% and recall rate of 93.98%.

Work Experience

Applied Scientist Intern, Amazon Sep 2025 - Dec 2025

Research with projects of evaluation and optimization of LLM agents *San Diego, U.S.*

- Improved natural language optimization of LLMs on the GEPA framework and Amazon Bedrock platform.
- Implemented prompt learning rate and token usage controls to optimize training efficiency cost.

Research Intern, Ping An Technology May 2025 - Aug 2025

Research Internship about LLM Probing on reference-free evaluation tasks *Shenzhen, China*

- Published **ICLR 2026**: "Rethinking LLM-as-a-Judge: Representation-as-a-Judge with Small Language Models via Semantic Capacity Asymmetry".
- Probed small LMs' layer representations evaluation scores across 5 aspects on {question, answer} pairs.

Machine Learning Engineer Intern, MEDA AI May 2022 - Aug 2022

Internship with Startup Half Moon Tech to work on the Meda Metaverse project *Remote, U.S.*

- Improved the text-to-speech (TTS) based on the Tacotron model and built server to train over 10M steps.
- Assisted in constructing 3D character model and got average accuracy over 85% on facial attributes classification task on dataset CelebA.

Software Engineering Intern, Pactera Technology May 2021 - Aug 2021

Design and Development of an Intelligent Customer Service System *Wuhan, China*

- Conducted field surveys over 100 customers, assisted designing database E-R model containing 34 tables.
- Developed online semantic analysis system using Baidu voice recognition API with 80% code coverage.

Skills

Programming Language/Platform: Python (main), C, C++, JAVA, SQL, Linux, GIT, AWS, Matlab.

Tools: Anaconda, Pytorch, Huggingface, Numpy, Pandas, Scikit-learn, Keras, Spacy, NLTK, Rasa_nlu, Word2Vec.